

Statistical Analysis Plan

Efficacy Analysis Plan - Xanomeline-Tropium Phase 3

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1. Primary Analysis

The primary endpoint is the change from baseline to Week 24 in the Alzheimer's Disease Assessment Scale-Cognitive Subscale (ADAS-Cog) total score. The primary analysis will be conducted using a Mixed Model for Repeated Measures (MMRM) approach. The model will include fixed effects for treatment group, visit week, treatment-by-visit interaction, and stratification factors (disease severity and concomitant cholinesterase inhibitor use). Baseline ADAS-Cog score will be included as a continuous covariate. An unstructured covariance matrix will be used to model the within-patient correlation. Kenward-Roger approximation will be used for denominator degrees of freedom.

2. Sensitivity Analyses

Several sensitivity analyses will be performed: (1) Pattern-mixture model to assess the impact of missing data under non-ignorable missingness assumptions; (2) Analysis using multiple imputation under missing at random (MAR) assumption; (3) Analysis of completers only (patients who complete Week 24 assessment); (4) Analysis using tipping-point approach to determine how much the treatment effect would need to shift to lose statistical significance.

3. Subgroup Analyses

Pre-specified subgroup analyses will be performed for the following factors: age group (50-64, 65-74, 75-85 years), disease severity (mild MMSE 20-26 vs. moderate MMSE 14-19), APOE4 carrier status (carrier vs. non-carrier), sex (male vs. female), and concomitant cholinesterase inhibitor use (yes vs. no). These analyses will be performed using the same MMRM model with the addition of subgroup-by-treatment interaction terms.

4. Multiplicity Adjustment

To control the family-wise Type I error rate at 0.05 across the primary and key secondary endpoints, a hierarchical testing procedure will be employed. The testing sequence is: (1) ADAS-Cog at Week 24 (primary), (2) CIBIC+ at Week 24, (3) ADCS-ADL at Week 24, (4) MMSE at Week 24. Each endpoint will be tested at the two-sided 0.05 level only if all preceding endpoints in the hierarchy are significant.